Unsupervised Learning and Feature Engineering

This notebook presents the unsupervised learning and feature engineering component of the final project pipeline.

The goal of this notebook is to identify meaningful trip patterns in the NYC TLC High Volume For-Hire Vehicles (HVFHV) dataset using clustering, dimensionality reduction, and feature engineering.

This notebook focuses on:

1. feature engineering for clustering,
2. preprocessing and cluster evaluation,
3. PCA-based visualization,
4. cluster interpretation and large-scale validation.

The results presented here complement the EDA, baseline modeling, and advanced supervised learning components developed in the other project notebooks.

In [ ]:
```
from pathlib import Path
import warnings

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, calinski_harabasz_score, davie

warnings.filterwarnings("ignore")

pd.set_option("display.max_columns", None)
pd.set_option("display.float_format", lambda x: f"{x:,.4f}")

sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (8, 5)

RANDOM_STATE = 42

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

Data Loading and Environment Setup

This notebook is designed to support the final unsupervised learning workflow at multiple dataset scales.

To balance reproducibility, computational feasibility, and analytical rigor, the workflow uses different dataset modes for different stages of the analysis:

- `100k sample dataset` : for lightweight testing and debugging,
- `5M dataset` : for the main unsupervised workflow,
- `300k analysis subset drawn from the 5M dataset` : for computationally expensive cluster-selection steps,
- `full cleaned dataset` : for final validation and interpretation.

This design allows the notebook to remain computationally practical while ensuring that the final conclusions are supported by large-scale data.

In [ ]:

```
# ================================
# Dataset Configuration
# ================================
DATA_MODE = "eda_5m"   # options: "sample", "eda_5m", "full"

# Project path
PROJECT_DIR = Path.cwd()

FULL_DATA_PATH = PROJECT_DIR / "tripdata_clean_full.parquet"
EDA_5M_PATH = PROJECT_DIR / "tripdata_clean_eda_5M.parquet"
SAMPLE_DATA_PATH = PROJECT_DIR / "tripdata_sample_100k.parquet"

print("Current project directory:", PROJECT_DIR)
print("Full dataset exists:", FULL_DATA_PATH.exists())
print("5M dataset exists:", EDA_5M_PATH.exists())
print("100k sample dataset exists:", SAMPLE_DATA_PATH.exists())

# ================================
# Select dataset
# ================================
if DATA_MODE == "full":
    DATA_PATH = FULL_DATA_PATH
    dataset_name = "full cleaned dataset"
elif DATA_MODE == "eda_5m":
    DATA_PATH = EDA_5M_PATH
    dataset_name = "5M EDA dataset"
elif DATA_MODE == "sample":
    DATA_PATH = SAMPLE_DATA_PATH
    dataset_name = "100k sample dataset"
else:
    raise ValueError("DATA_MODE must be one of: 'sample', 'eda_5m', or 'full

print(f"\nLoading {dataset_name}...")
print("Selected file:", DATA_PATH)

# Load the core columns required for clustering and feature engineering
columns_needed = [
```

```
    "hvfhs_license_num",
    "pickup_datetime",
    "trip_miles",
    "trip_time",
    "driver_pay"
]

df = pd.read_parquet(DATA_PATH, columns=columns_needed)

print("Loaded dataset shape:", df.shape)
print("Current DATA_MODE:", DATA_MODE)

df.head()
```

```
Current project directory: /Users/xsw/Desktop/MyProject/COMP333-Group Projec
t
Full dataset exists: True
5M dataset exists: True
100k sample dataset exists: True

Loading 5M EDA dataset...
Selected file: /Users/xsw/Desktop/MyProject/COMP333-Group Project/tripdata_c
lean_eda_5M.parquet
Loaded dataset shape: (5000000, 5)
Current DATA_MODE: eda_5m
```

Out[ ]:

	hvfhs_license_num	pickup_datetime	trip_miles	trip_time	driver_pay
0	HV0003	2026-01-27 12:39:31	1.5900	1455	18.9100
1	HV0003	2026-01-01 16:42:50	1.1400	594	7.9300
2	HV0005	2026-01-18 16:20:38	3.7330	1991	26.5500
3	HV0003	2026-01-09 07:36:57	9.9000	1731	31.3000
4	HV0005	2025-12-16 11:51:36	2.7730	1413	19.4200

Initial Inspection and Feature Planning

Before constructing clustering features, the selected dataset is inspected to confirm the structure, data types, and completeness of the variables used in the unsupervised workflow.

This step is important for two reasons:

- it verifies that the selected columns are suitable for feature engineering and clustering,
- it helps identify distributional properties that may affect later preprocessing and interpretation.

The unsupervised analysis is designed to capture both trip behavior and temporal structure. The planned feature space therefore includes:

- trip-level variables such as `trip_miles`, `trip_time`, and `driver_pay`,
- time-based variables derived from `pickup_datetime`,
- engineered features intended to better represent trip behavior.

In [ ]:
```
print("Inspection dataset mode:", DATA_MODE)
print("Dataset shape:", df.shape)

print("\nColumn data types:")
print(df.dtypes)

print("\nMissing values by column:")
print(df.isnull().sum())

print("\nSummary statistics:")
display(df[["trip_miles", "trip_time", "driver_pay"]].describe())

print("\nUnique provider codes:")
print(df["hvfhs_license_num"].value_counts())

print("\nPickup datetime range:")
print("Min:", df["pickup_datetime"].min())
print("Max:", df["pickup_datetime"].max())
```

```
Inspection dataset mode: eda_5m
Dataset shape: (5000000, 5)

Column data types:
hvfhs_license_num            object
pickup_datetime      datetime64[us]
trip_miles                  float32
trip_time                     int32
driver_pay                  float32
dtype: object

Missing values by column:
hvfhs_license_num    0
pickup_datetime      0
trip_miles           0
trip_time            0
driver_pay           0
dtype: int64

Summary statistics:
```

	trip_miles	trip_time	driver_pay
count	5,000,000.0000	5,000,000.0000	5,000,000.0000
mean	4.7471	1,162.9038	20.6265
std	5.6469	841.3147	17.8669
min	0.0030	1.0000	-24.7000
25%	1.4300	575.0000	9.0800
50%	2.7520	940.0000	15.4600
75%	5.8900	1,502.0000	26.1500
max	343.6000	35,715.0000	1,197.2700

```
Unique provider codes:
hvfhs_license_num
HV0003    3626656
HV0005    1373344
Name: count, dtype: int64

Pickup datetime range:
Min: 2025-12-01 00:00:00
Max: 2026-01-31 23:59:59
```

Feature Engineering for Clustering

To improve the clustering analysis, additional features are constructed from the cleaned trip records.

Feature engineering is especially important in unsupervised learning because the discovered cluster structure depends directly on how similarity is defined in the feature space. Using only raw variables such as trip distance and trip duration may overlook meaningful behavioral patterns.

In this notebook, two main types of engineered features are used:

- time-based features derived from `pickup_datetime`,
- a domain-specific behavioral feature, `speed_mph`, to better capture trip efficiency and movement characteristics.

These engineered features make the unsupervised workflow more informative and help support later interpretation of the resulting trip patterns.

In [ ]:
```
# Time-based features
df["pickup_hour"] = df["pickup_datetime"].dt.hour.astype("int8")
df["pickup_dayofweek"] = df["pickup_datetime"].dt.dayofweek.astype("int8")

# Domain-specific feature: average speed in miles per hour
df["speed_mph"] = (df["trip_miles"] / (df["trip_time"] / 3600)).astype("floa
```

```
feature_cols_preview = [
    "pickup_datetime",
    "pickup_hour",
    "pickup_dayofweek",
    "trip_miles",
    "trip_time",
    "driver_pay",
    "speed_mph"
]

print(f"Preview of engineered features on {dataset_name}:")
display(df[feature_cols_preview].head())

print("\nSummary statistics for engineered features:")
display(df[["pickup_hour", "pickup_dayofweek", "speed_mph"]].describe())
```

```
Preview of engineered features on 5M EDA dataset:
```

	pickup_datetime	pickup_hour	pickup_dayofweek	trip_miles	trip_time	driver_pay
0	2026-01-27 12:39:31	12	1	1.5900	1455	18.9100
1	2026-01-01 16:42:50	16	3	1.1400	594	7.9300
2	2026-01-18 16:20:38	16	6	3.7330	1991	26.5500
3	2026-01-09 07:36:57	7	4	9.9000	1731	31.3000
4	2025-12-16 11:51:36	11	1	2.7730	1413	19.4200

```
Summary statistics for engineered features:
```

	pickup_hour	pickup_dayofweek	speed_mph
count	5,000,000.0000	5,000,000.0000	5,000,000.0000
mean	13.4033	3.0579	13.0746
std	6.3907	1.9282	7.0819
min	0.0000	0.0000	0.0039
25%	9.0000	1.0000	8.3208
50%	14.0000	3.0000	11.0225
75%	19.0000	5.0000	15.8411
max	23.0000	6.0000	661.3260

Preprocessing for Clustering

Before applying PCA and KMeans clustering, the engineered features must be preprocessed carefully.

This step is necessary because clustering algorithms are highly sensitive to feature scale, skewness, and extreme observations. In the current dataset, variables such as `trip_miles`, `driver_pay`, and `speed_mph` show substantial right-skewness and contain extreme values that could otherwise dominate the clustering structure.

To improve clustering stability and interpretability, the preprocessing workflow includes:

- filtering implausible or extreme values in `speed_mph`,
- applying log transformations to strongly right-skewed variables,
- defining the final clustering feature set,
- removing rows with missing values in the clustering features.

These steps make the clustering pipeline more robust and help ensure that the final cluster structure reflects meaningful trip patterns rather than artifacts of raw scale differences.

In [ ]:
```
# ------------------------------
# Remove extreme speed values using the 99th percentile
# ------------------------------
initial_rows = len(df)

speed_upper_bound = df["speed_mph"].quantile(0.99)
df = df[(df["speed_mph"] > 0) & (df["speed_mph"] <= speed_upper_bound)]

print("Preprocessing dataset mode:", DATA_MODE)
print("Rows before speed filtering:", initial_rows)
print("Rows after speed filtering :", len(df))
print("Rows removed               :", initial_rows - len(df))
print("99th percentile of speed_mph:", speed_upper_bound)

# ------------------------------
# Log-transform right-skewed variables
# ------------------------------
df["log_trip_miles"] = np.log1p(df["trip_miles"]).astype("float32")
df["log_driver_pay"] = np.log1p(df["driver_pay"]).astype("float32")
df["log_speed_mph"] = np.log1p(df["speed_mph"]).astype("float32")

# ------------------------------
# Final feature set for clustering
# ------------------------------
cluster_features = [
    "log_trip_miles",
    "trip_time",
    "log_driver_pay",
    "pickup_hour",
    "pickup_dayofweek",
    "log_speed_mph"
]
```

```python
# Remove rows with missing values in the final clustering features
before_dropna = len(df)
df = df.dropna(subset=cluster_features)

print("\nRows before dropping NaNs in clustering features:", before_dropna)
print("Rows after dropping NaNs in clustering features :", len(df))
print("Rows removed due to NaNs                        :", before_dropna - l

print("\nSelected clustering features:")
print(cluster_features)

print("\nSummary statistics after preprocessing:")
display(df[cluster_features].describe())
```

```
Preprocessing dataset mode: eda_5m
Rows before speed filtering: 5000000
Rows after speed filtering : 4950000
Rows removed               : 50000
99th percentile of speed_mph: 36.44776355743404

Rows before dropping NaNs in clustering features: 4950000
Rows after dropping NaNs in clustering features : 4949991
Rows removed due to NaNs                        : 9

Selected clustering features:
['log_trip_miles', 'trip_time', 'log_driver_pay', 'pickup_hour', 'pickup_day
ofweek', 'log_speed_mph']

Summary statistics after preprocessing:
```

	log_trip_miles	trip_time	log_driver_pay	pickup_hour	pickup_dayofweel
count	4,949,991.0000	4,949,991.0000	4,949,991.0000	4,949,991.0000	4,949,991.000
mean	1.4392	1,149.9248	2.8129	13.4221	3.057
std	0.6914	824.6453	0.6777	6.3691	1.927
min	0.0030	1.0000	0.0000	0.0000	0.000
25%	0.8838	572.0000	2.3046	9.0000	1.000
50%	1.3135	933.0000	2.7918	14.0000	3.000
75%	1.9066	1,485.0000	3.2862	19.0000	5.000
max	5.4018	35,715.0000	6.9515	23.0000	6.000

Choosing the Number of Clusters

To determine a reasonable value of `k`, multiple candidate cluster counts are compared using several evaluation metrics.

Because full cluster-search evaluation is computationally expensive on millions of observations, this step is performed on a large analysis subset drawn from the 5M

dataset. This preserves the large-scale character of the workflow while keeping the model-selection stage computationally feasible.

The clustering quality is evaluated using:

- inertia (within-cluster sum of squares),
- silhouette score,
- Calinski-Harabasz score,
- Davies-Bouldin score.

Together, these metrics provide a more comprehensive evaluation of cluster compactness, separation, and interpretability than any single metric alone. This section therefore serves as the main model-selection and evaluation stage of the unsupervised workflow.

In [ ]:
```
# Create a large analysis subset for cluster selection
analysis_sample_size = 300_000

if len(df) > analysis_sample_size:
    df_ksearch = df.sample(n=analysis_sample_size, random_state=RANDOM_STATE
else:
    df_ksearch = df

print("Cluster-selection dataset shape:", df_ksearch.shape)

# Standardize the clustering features on the analysis subset
X_ksearch = df_ksearch[cluster_features]

scaler_ksearch = StandardScaler()
X_ksearch_scaled = scaler_ksearch.fit_transform(X_ksearch)

# Evaluate candidate values of k
k_values = range(2, 9)
results = []

for k in k_values:
    print(f"Evaluating k = {k}...")

    kmeans = KMeans(
        n_clusters=k,
        random_state=RANDOM_STATE,
        n_init=10
    )

    cluster_labels = kmeans.fit_predict(X_ksearch_scaled)

    inertia = kmeans.inertia_
    silhouette = silhouette_score(X_ksearch_scaled, cluster_labels)
    calinski = calinski_harabasz_score(X_ksearch_scaled, cluster_labels)
    davies = davies_bouldin_score(X_ksearch_scaled, cluster_labels)

    results.append({
        "k": k,
```

```
        "inertia": inertia,
        "silhouette_score": silhouette,
        "calinski_harabasz_score": calinski,
        "davies_bouldin_score": davies
    })

cluster_eval_df = pd.DataFrame(results)

print("\nCluster evaluation results:")
display(cluster_eval_df)
```

```
Cluster-selection dataset shape: (300000, 11)
Evaluating k = 2...
Evaluating k = 3...
Evaluating k = 4...
Evaluating k = 5...
Evaluating k = 6...
Evaluating k = 7...
Evaluating k = 8...

Cluster evaluation results:
```

	k	inertia	silhouette_score	calinski_harabasz_score	davies_bouldin_score
0	2	1,176,490.1728	0.3153	158,991.4277	1.2823
1	3	1,024,493.0600	0.2111	113,544.4932	1.5835
2	4	904,342.7849	0.2025	99,038.8669	1.4664
3	5	800,624.0949	0.2057	93,617.5420	1.4687
4	6	730,344.4822	0.2013	87,874.1780	1.3965
5	7	670,120.7974	0.2072	84,302.7461	1.3379
6	8	625,892.5988	0.2076	80,393.5722	1.2717

Scaling and Principal Component Analysis (PCA)

Before fitting the clustering model, the selected features are standardized so that variables measured on different scales contribute more comparably to the analysis.

Principal Component Analysis (PCA) is then applied for two main purposes:

- to reduce the dimensionality of the clustering feature space,
- to support visualization of the latent structure of the trips in a lower-dimensional representation.

PCA is appropriate here because the clustering feature set contains multiple correlated numerical variables. A lower-dimensional view helps summarize the dominant structure in the data and provides a clearer visual basis for interpreting the final clustering results.

In [ ]:
```
# ------------------------------
# Standardize clustering features on the analysis subset
# ------------------------------
X_ksearch = df_ksearch[cluster_features]

scaler_ksearch = StandardScaler()
X_ksearch_scaled = scaler_ksearch.fit_transform(X_ksearch)

print("Scaled feature matrix shape:", X_ksearch_scaled.shape)

# ------------------------------
# Apply PCA
# ------------------------------
pca = PCA()
X_pca = pca.fit_transform(X_ksearch_scaled)

explained_variance = pca.explained_variance_ratio_
cumulative_variance = explained_variance.cumsum()

pca_summary = pd.DataFrame({
    "PC": [f"PC{i+1}" for i in range(len(explained_variance))],
    "Explained Variance Ratio": explained_variance,
    "Cumulative Variance": cumulative_variance
})

print("\nPCA summary:")
display(pca_summary)
```

```
Scaled feature matrix shape: (300000, 6)

PCA summary:
```

	PC	Explained Variance Ratio	Cumulative Variance
0	PC1	0.5047	0.5047
1	PC2	0.1830	0.6877
2	PC3	0.1624	0.8501
3	PC4	0.1275	0.9776
4	PC5	0.0192	0.9967
5	PC6	0.0033	1.0000

PCA Visualization

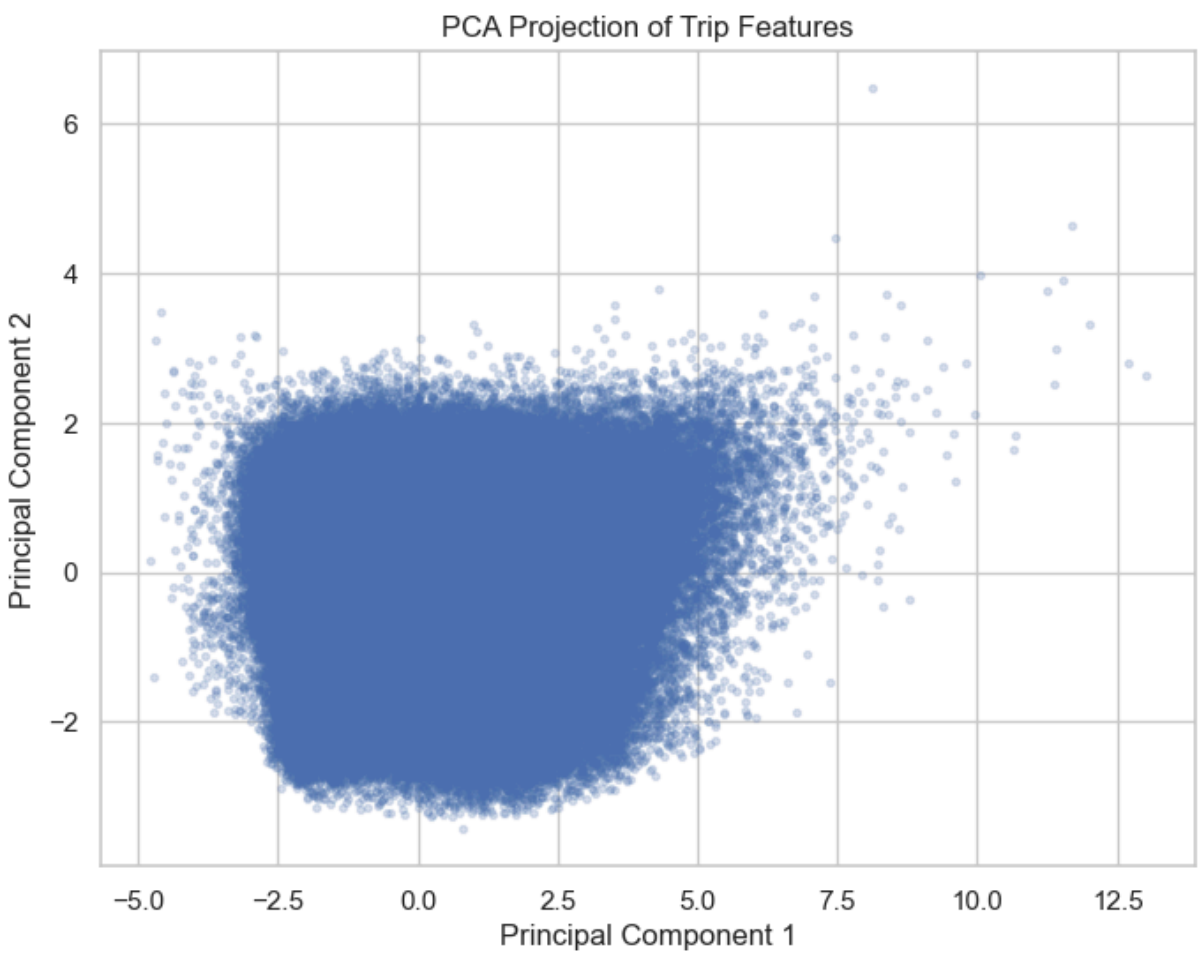
The first two principal components are visualized to examine the overall structure of the clustering feature space.

Although PCA itself does not produce clusters, it provides a useful lower-dimensional summary of the data. This makes it easier to assess whether meaningful structure is

present and also creates a consistent visual frame for displaying the final cluster assignments after KMeans is applied.

In this notebook, PCA is used primarily as an interpretive and visualization tool rather than as the clustering method itself.

In [ ]:
```
# Create a DataFrame for the first two principal components
pca_df = pd.DataFrame({
    "PC1": X_pca[:, 0],
    "PC2": X_pca[:, 1]
})

# Scatter plot of the first two principal components
plt.figure(figsize=(8, 6))
plt.scatter(
    pca_df["PC1"],
    pca_df["PC2"],
    alpha=0.2,
    s=8
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA Projection of Trip Features")
plt.show()
```



Cluster Evaluation Plots

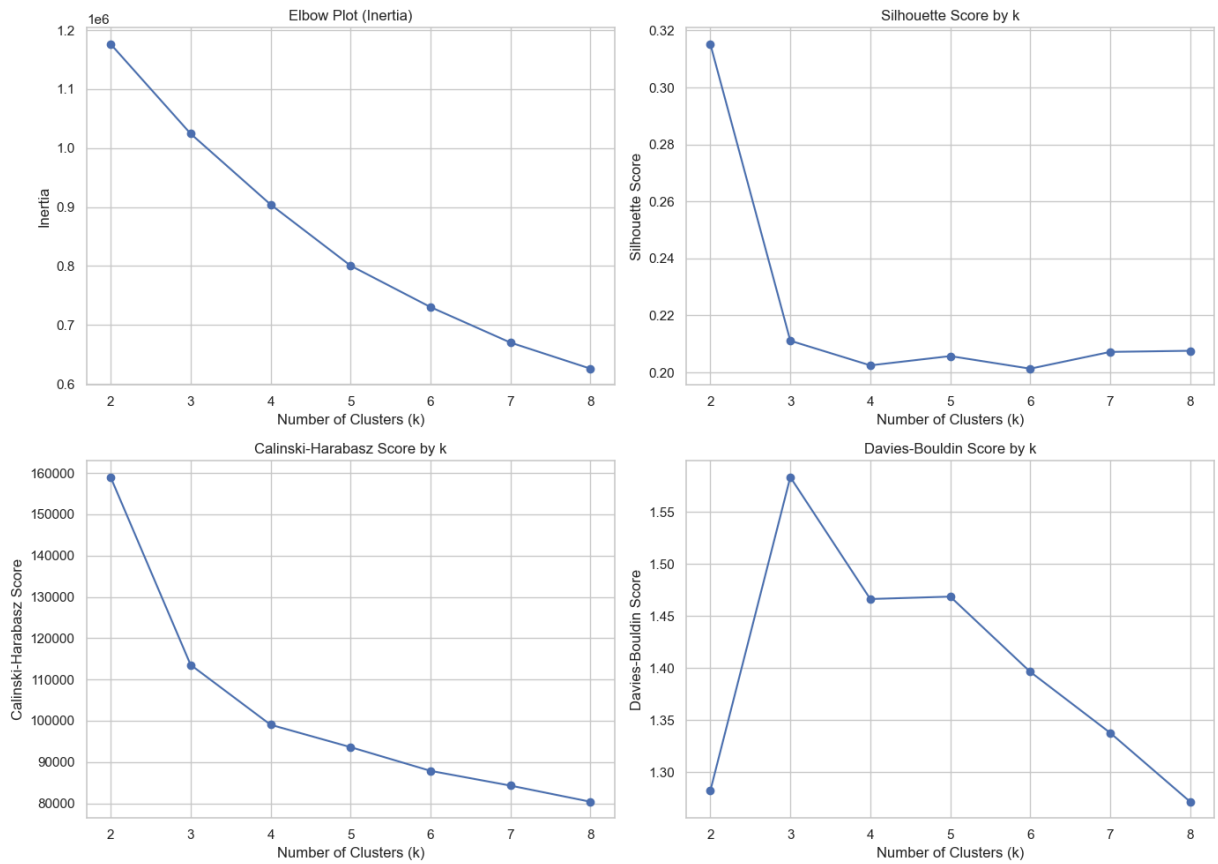
To compare candidate values of `k` more clearly, the clustering evaluation metrics are visualized.

These plots provide a compact summary of how the clustering solution changes as the number of clusters increases. In particular, they help show:

- how within-cluster variation changes across `k`,
- whether the silhouette, Calinski-Harabasz, and Davies-Bouldin metrics support a more interpretable solution,
- whether the selected `k` is justified by a combination of quantitative evidence rather than by arbitrary choice.

These visualizations form an important part of the comprehensive evaluation of the unsupervised workflow.

In [ ]:
```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Inertia
axes[0, 0].plot(cluster_eval_df["k"], cluster_eval_df["inertia"], marker="o"
axes[0, 0].set_title("Elbow Plot (Inertia)")
axes[0, 0].set_xlabel("Number of Clusters (k)")
axes[0, 0].set_ylabel("Inertia")

# Silhouette Score
axes[0, 1].plot(cluster_eval_df["k"], cluster_eval_df["silhouette_score"], m
axes[0, 1].set_title("Silhouette Score by k")
axes[0, 1].set_xlabel("Number of Clusters (k)")
axes[0, 1].set_ylabel("Silhouette Score")

# Calinski-Harabasz Score
axes[1, 0].plot(cluster_eval_df["k"], cluster_eval_df["calinski_harabasz_sco
axes[1, 0].set_title("Calinski-Harabasz Score by k")
axes[1, 0].set_xlabel("Number of Clusters (k)")
axes[1, 0].set_ylabel("Calinski-Harabasz Score")

# Davies-Bouldin Score
axes[1, 1].plot(cluster_eval_df["k"], cluster_eval_df["davies_bouldin_score"
axes[1, 1].set_title("Davies-Bouldin Score by k")
axes[1, 1].set_xlabel("Number of Clusters (k)")
axes[1, 1].set_ylabel("Davies-Bouldin Score")

plt.tight_layout()
plt.show()
```



Selecting the Final Number of Clusters and Training KMeans

Based on the cluster evaluation metrics, a final value of `k` is selected for the KMeans model.

The selected solution is intended to balance quantitative cluster quality with interpretability. While larger values of `k` may produce more granular segmentation, the final choice should also remain stable, meaningful, and explainable in the context of ride-share trip behavior.

After selecting the final value of `k`, the KMeans model is trained on the large analysis subset. The resulting cluster assignments are then used for visualization, profiling, and interpretation in the sections that follow.

In [ ]:
```
# ------------------------------
# Train final KMeans model on the analysis subset
# ------------------------------
final_k = 2

final_kmeans = KMeans(
    n_clusters=final_k,
    random_state=RANDOM_STATE,
    n_init=10
)
```

```
df_ksearch["cluster"] = final_kmeans.fit_predict(X_ksearch_scaled)

print("Final number of clusters:", final_k)
print("\nCluster counts:")
print(df_ksearch["cluster"].value_counts().sort_index())
```

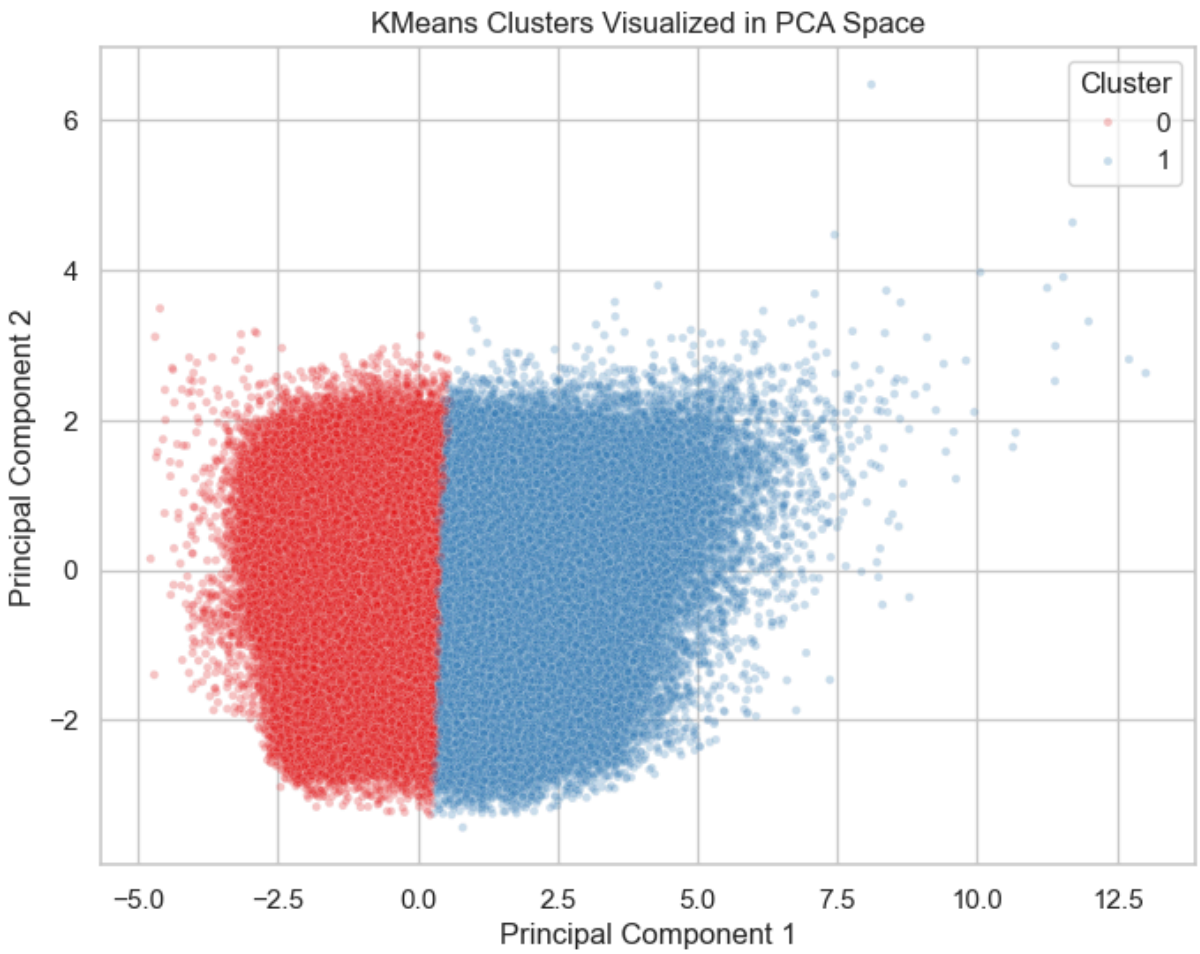
```
Final number of clusters: 2

Cluster counts:
cluster
0    190563
1    109437
Name: count, dtype: int64
```

PCA Visualization of the Final Clusters

After selecting the final value of `k`, the resulting cluster assignments are visualized in the space of the first two principal components.

This plot does not replace the quantitative evaluation metrics, but it provides an intuitive visual summary of how the final clusters are distributed in a lower-dimensional representation of the feature space. It is especially useful for checking whether the final solution exhibits visible separation and whether the identified patterns appear coherent at a high level.

In [ ]:
```
# Add cluster labels to the PCA DataFrame
pca_df["cluster"] = df_ksearch["cluster"].values

# Visualize the final clusters in PCA space
plt.figure(figsize=(8, 6))

sns.scatterplot(
    data=pca_df,
    x="PC1",
    y="PC2",
    hue="cluster",
    palette="Set1",
    alpha=0.25,
    s=12
)

plt.title("KMeans Clusters Visualized in PCA Space")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend(title="Cluster")
plt.show()
```



Cluster Profiling

To interpret the meaning of the final clusters, summary statistics are computed for each cluster.

Cluster profiling is essential because unsupervised learning does not provide predefined labels. Instead, the cluster assignments must be translated into interpretable trip patterns by comparing the groups across relevant variables.

In this notebook, the clusters are profiled using:

- trip behavior variables such as `trip_miles`, `trip_time`, `driver_pay`, and `speed_mph`,
- temporal variables such as `pickup_hour` and `pickup_dayofweek`.

This step helps determine whether the final clustering solution corresponds to meaningful ride-share trip patterns rather than purely mathematical partitions.

In [ ]:
```
cluster_profile = df_ksearch.groupby("cluster")[
    [
        "trip_miles",
        "trip_time",
```

```
        "driver_pay",
        "speed_mph",
        "pickup_hour",
        "pickup_dayofweek"
    ]
].agg(["mean", "median", "std"])

print("Cluster profile summary:")
display(cluster_profile)
```

```
Cluster profile summary:
```

	trip_miles			trip_time			drive	
	mean	median	std	mean	median	std	mean	median
cluster								
0	1.8778	1.6900	0.9702	725.5330	663.0000	363.5264	11.5667	10.5400
1	9.0763	7.4000	5.4369	1,884.4227	1,682.0000	875.8771	34.8692	29.9000

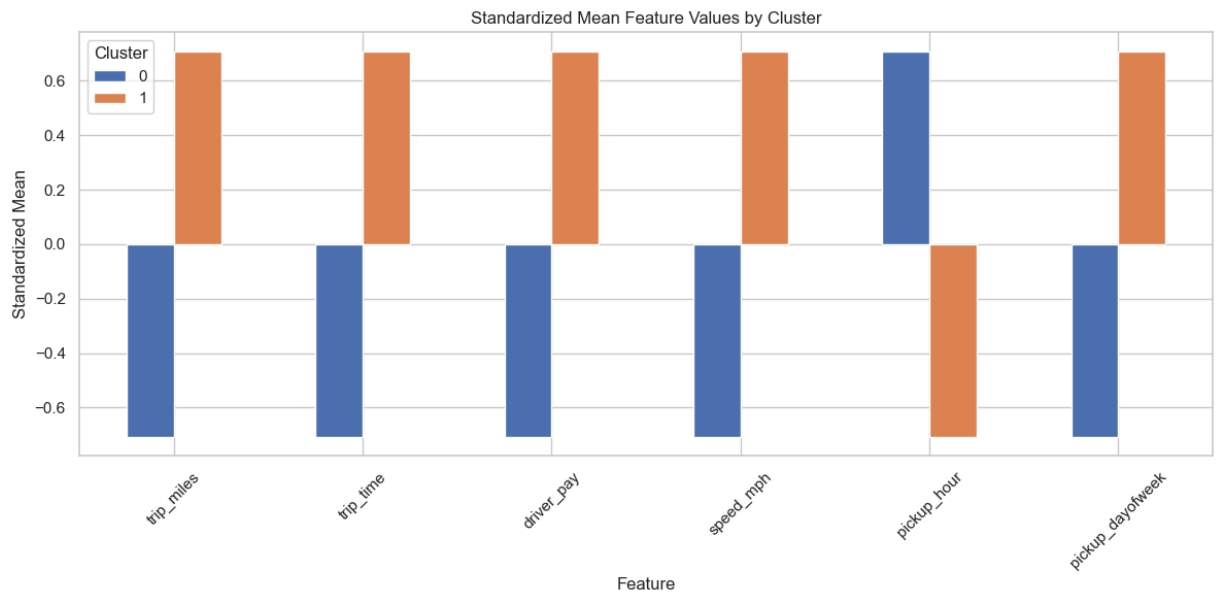
Standardized Cluster Profile Visualization

Because the clustering variables are measured on different scales, a direct comparison of raw cluster means can be misleading.

To make the differences between clusters easier to interpret, the feature means are standardized before visualization. This allows the plot to emphasize relative patterns across variables rather than their original magnitudes.

As a result, the standardized profile visualization helps clarify which characteristics distinguish the clusters most strongly in comparative terms.

In [ ]:
```
cluster_profile_means = df_ksearch.groupby("cluster")[
    ["trip_miles", "trip_time", "driver_pay", "speed_mph", "pickup_hour", "p
].mean()

# Standardize the cluster means column-wise
cluster_profile_standardized = (
    cluster_profile_means - cluster_profile_means.mean()
) / cluster_profile_means.std()

cluster_profile_standardized.T.plot(kind="bar", figsize=(12, 6))

plt.title("Standardized Mean Feature Values by Cluster")
plt.xlabel("Feature")
plt.ylabel("Standardized Mean")
plt.xticks(rotation=45)
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```



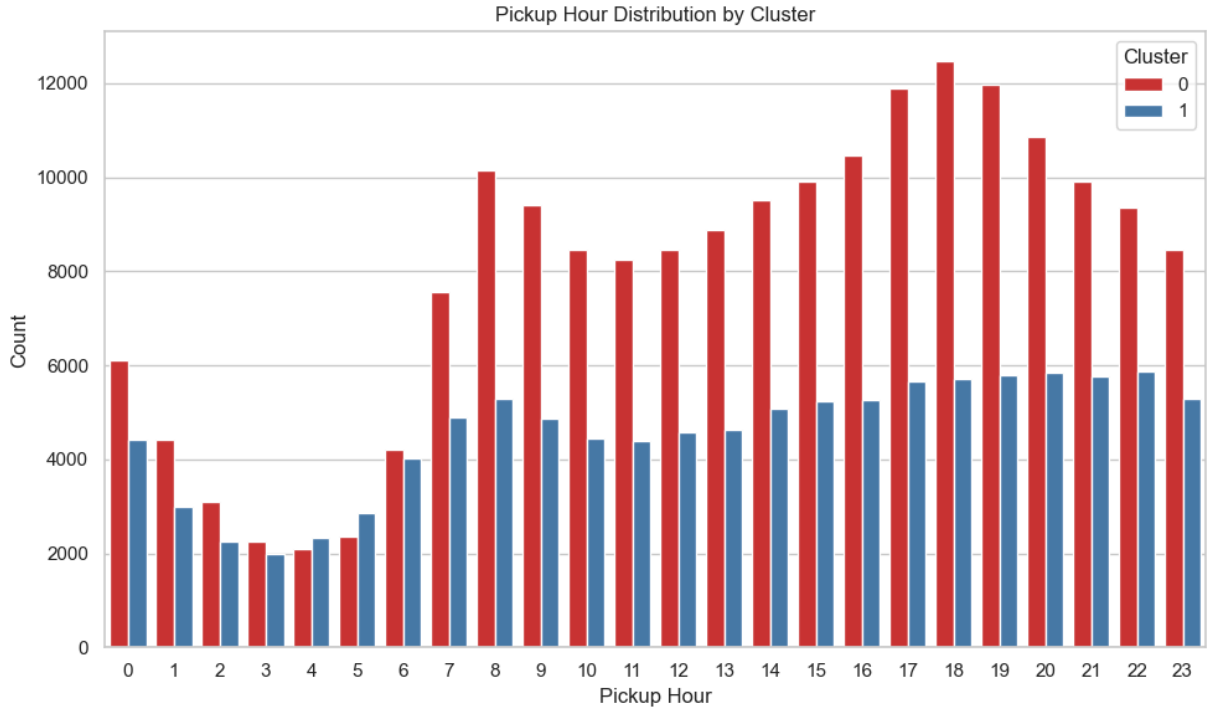
Time Patterns by Cluster

To better understand the role of temporal structure in the clustering solution, the distribution of pickup hour is compared across clusters.

This analysis helps assess whether the final trip patterns are strongly differentiated by time of day or whether temporal variables play a more secondary role relative to trip behavior variables such as distance, duration, and driver pay.

The comparison therefore supports a more complete interpretation of the cluster structure.

In [ ]:
```
plt.figure(figsize=(10, 6))

sns.countplot(
    data=df_ksearch,
    x="pickup_hour",
    hue="cluster",
    palette="Set1"
)

plt.title("Pickup Hour Distribution by Cluster")
plt.xlabel("Pickup Hour")
plt.ylabel("Count")
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```



Provider Distribution by Cluster

To further interpret the final clustering solution, the distribution of service providers is examined across clusters.

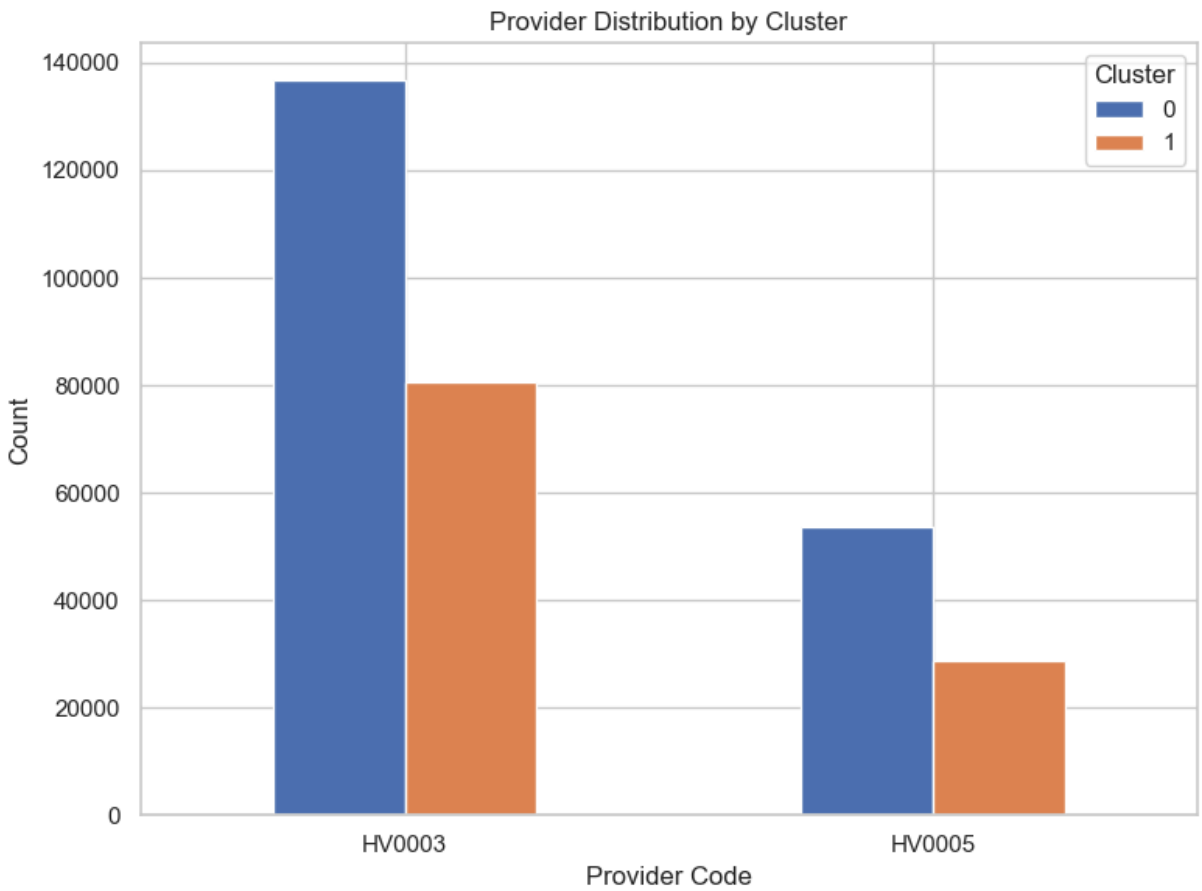
Although `hvfhs_license_num` is not directly used as a clustering feature, it provides an important external grouping variable for interpretation. In particular, this comparison helps assess whether the clusters reflect broad trip behavior patterns shared across providers or whether the solution is simply separating the platforms themselves.

This step therefore adds contextual interpretation to the unsupervised workflow without changing the clustering model.

In [ ]:
```
provider_cluster_counts = pd.crosstab(df_ksearch["hvfhs_license_num"], df_ks

print("Provider distribution by cluster:")
display(provider_cluster_counts)

provider_cluster_counts.plot(kind="bar", figsize=(8, 6))

plt.title("Provider Distribution by Cluster")
plt.xlabel("Provider Code")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```

```
Provider distribution by cluster:
```

cluster	0	1
hvfhs_license_num		
HV0003	136790	80548
HV0005	53773	28889



Full-Dataset Validation

To assess the robustness of the clustering workflow, the same preprocessing pipeline is applied to the full cleaned dataset and the clustering model learned from the 5M analysis workflow is used to assign cluster labels at full scale.

This step is important because the final analytical claims should not depend only on the 5M workflow or on the analysis subset used during cluster selection. Instead, the full-dataset validation checks whether the main structure identified earlier remains consistent when the model is applied to the complete cleaned dataset.

In this way, the validation stage serves as the main robustness check for the unsupervised component of the final project pipeline.

In [ ]:
```
# Reload the full cleaned dataset for final validation
full_columns_needed = [
    "hvfhs_license_num",
```

```
    "pickup_datetime",
    "trip_miles",
    "trip_time",
    "driver_pay"
]

df = pd.read_parquet(FULL_DATA_PATH, columns=full_columns_needed)

print("Reloaded full dataset shape:", df.shape)

# Re-apply feature engineering to the full dataset
df["pickup_hour"] = df["pickup_datetime"].dt.hour.astype("int8")
df["pickup_dayofweek"] = df["pickup_datetime"].dt.dayofweek.astype("int8")
df["speed_mph"] = (df["trip_miles"] / (df["trip_time"] / 3600)).astype("floa

# Re-apply preprocessing to the full dataset
speed_upper_bound_full = df["speed_mph"].quantile(0.99)
df = df[(df["speed_mph"] > 0) & (df["speed_mph"] <= speed_upper_bound_full)]

df["log_trip_miles"] = np.log1p(df["trip_miles"]).astype("float32")
df["log_driver_pay"] = np.log1p(df["driver_pay"]).astype("float32")
df["log_speed_mph"] = np.log1p(df["speed_mph"]).astype("float32")

df = df.dropna(subset=cluster_features)

print("Full dataset shape after preprocessing:", df.shape)

# Apply the trained scaler and final KMeans model to the full dataset
X_full = df[cluster_features]
X_full_scaled = scaler_ksearch.transform(X_full)

df["cluster"] = final_kmeans.predict(X_full_scaled)

print("Full-dataset cluster counts:")
print(df["cluster"].value_counts().sort_index())
```

```
Reloaded full dataset shape: (43043669, 5)
Full dataset shape after preprocessing: (42613172, 11)
Full-dataset cluster counts:
cluster
0    27101389
1    15511783
Name: count, dtype: int64
```

Full-Dataset Cluster Profiling

After assigning cluster labels to the full cleaned dataset, the cluster profiles are summarized again to check whether the main trip-pattern differences observed in the 5M workflow remain consistent at full scale.

This step extends the interpretation of the clustering solution beyond the analysis subset and helps confirm whether the identified groups remain meaningful when applied to the full project dataset.

In [ ]:
```
cluster_profile_full = df.groupby("cluster")[
    [
        "trip_miles",
        "trip_time",
        "driver_pay",
        "speed_mph",
        "pickup_hour",
        "pickup_dayofweek"
    ]
].agg(["mean", "median", "std"])

print("Full-dataset cluster profile summary:")
display(cluster_profile_full)
```

```
Full-dataset cluster profile summary:
```

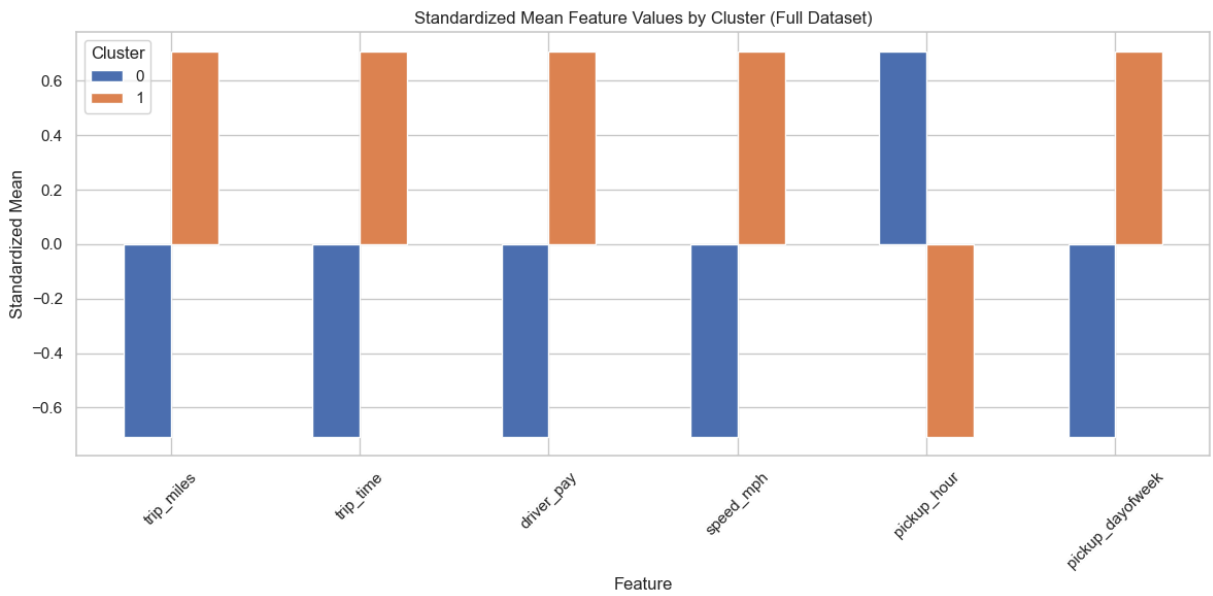
	trip_miles			trip_time			driver_		
	mean	median	std	mean	median	std	mean	median	
cluster									
0	1.8785	1.6900	0.9700	726.7635	665.0000	363.9702	11.5712	10.5500	5.
1	9.1072	7.4200	5.4732	1,890.2201	1,690.0000	883.5941	34.9718	29.9700	18.

Full-Dataset Standardized Cluster Profiles

To compare the full-dataset cluster profiles more clearly across variables with different scales, the cluster means are standardized before visualization.

As in the earlier profiling stage, this standardized comparison highlights the relative differences between clusters and helps show whether the same distinguishing structure remains visible when the workflow is applied to the full cleaned dataset.

In [ ]:
```
cluster_profile_means_full = df.groupby("cluster")[
    ["trip_miles", "trip_time", "driver_pay", "speed_mph", "pickup_hour", "p
].mean()

cluster_profile_standardized_full = (
    cluster_profile_means_full - cluster_profile_means_full.mean()
) / cluster_profile_means_full.std()

cluster_profile_standardized_full.T.plot(kind="bar", figsize=(12, 6))

plt.title("Standardized Mean Feature Values by Cluster (Full Dataset)")
plt.xlabel("Feature")
plt.ylabel("Standardized Mean")
plt.xticks(rotation=45)
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```

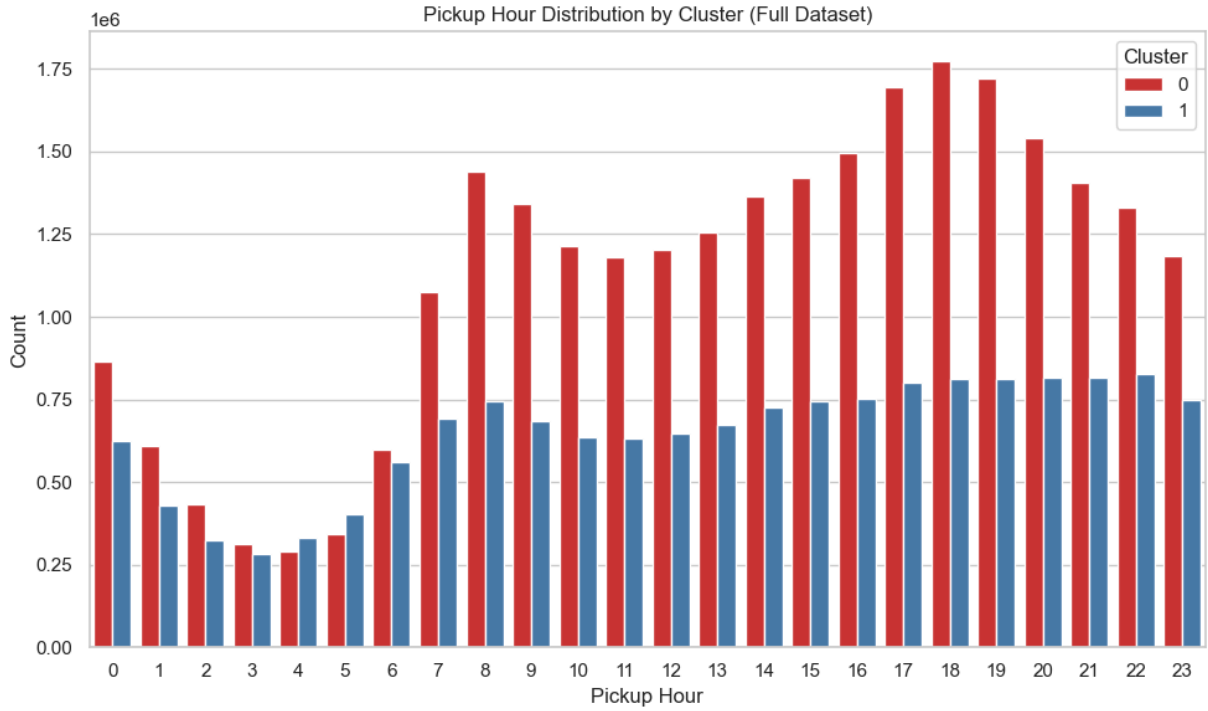


Full-Dataset Time Patterns by Cluster

The pickup-hour distribution is examined again on the full cleaned dataset to assess whether the temporal differences observed earlier remain consistent at full scale.

This comparison helps determine whether the role of temporal structure in the clustering solution is stable when moving from the 5M workflow to the full project dataset.

In [ ]:
```
plt.figure(figsize=(10, 6))

sns.countplot(
    data=df,
    x="pickup_hour",
    hue="cluster",
    palette="Set1"
)

plt.title("Pickup Hour Distribution by Cluster (Full Dataset)")
plt.xlabel("Pickup Hour")
plt.ylabel("Count")
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```



Full-Dataset Provider Distribution by Cluster

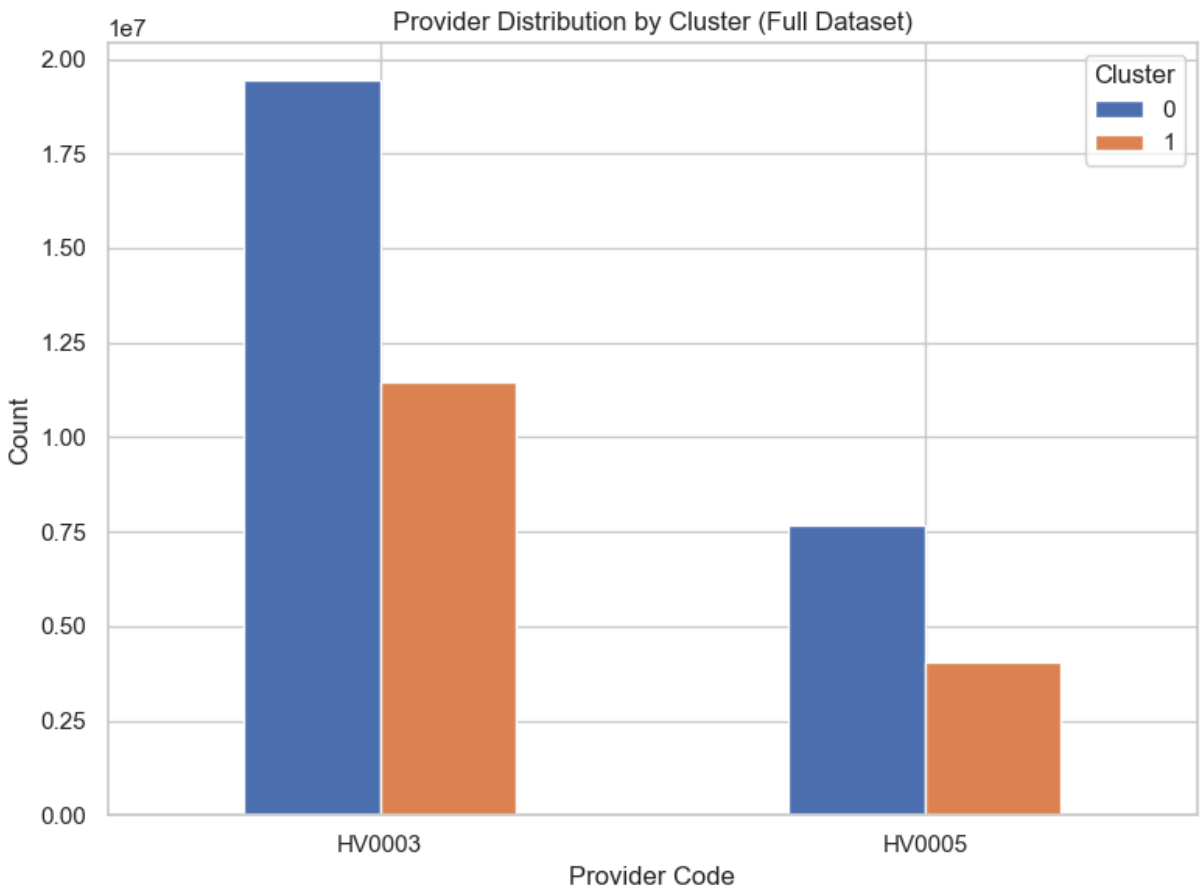
Finally, the distribution of service providers across clusters is checked again on the full cleaned dataset.

This step helps confirm whether the provider context observed in the earlier analysis remains consistent at full scale and whether the final clustering solution continues to reflect shared trip behavior patterns rather than simply separating providers.

In [ ]:
```
provider_cluster_counts_full = pd.crosstab(df["hvfhs_license_num"], df["clus

print("Full-dataset provider distribution by cluster:")
display(provider_cluster_counts_full)

provider_cluster_counts_full.plot(kind="bar", figsize=(8, 6))

plt.title("Provider Distribution by Cluster (Full Dataset)")
plt.xlabel("Provider Code")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```

```
Full-dataset provider distribution by cluster:
```

cluster	0	1
hvfhs_license_num		
HV0003	19445005	11456513
HV0005	7656384	4055270



Interpretation

The clustering analysis identifies two main trip patterns in the large-scale unsupervised workflow.

One cluster is characterized by longer trip distances, longer trip durations, higher driver pay, and higher average speed. This pattern is consistent with relatively larger and more efficient trips, which may include longer-distance rides or trips taking place under less congested conditions.

The other cluster is characterized by shorter trip distances, shorter trip durations, lower driver pay, and lower average speed. This pattern is more consistent with shorter and lower-paying trips, likely representing more local or slower urban rides.

The temporal variables contribute some additional structure, but they are not the dominant source of separation. The pickup-hour comparisons suggest that the final

cluster distinction is driven more strongly by trip behavior variables than by time of day alone.

The provider analysis also shows that both major providers appear across the identified clusters. This supports the interpretation that the clustering solution is not simply separating platforms, but rather capturing broader behavioral trip patterns that occur across providers.

The workflow is then validated on the full cleaned dataset. The full-dataset validation is important because it functions as a robustness check: it tests whether the main cluster differences observed in the 5M workflow remain visible when the same preprocessing and clustering logic are applied at full scale.

Overall, the results support the project objective by showing that HVFHV trips can be grouped into meaningful behavioral patterns using a combination of engineered trip-level and temporal features.

Conclusion

This notebook established the unsupervised learning and feature engineering component of the final project pipeline.

Using PCA, KMeans clustering, engineered features, and large-scale validation, the analysis identified an interpretable two-cluster structure in the HVFHV trip data. The final clusters differ mainly in trip distance, trip duration, driver pay, and average speed, while temporal variables play a more secondary role.

The clustering workflow was evaluated using multiple cluster-quality metrics and then validated on the full cleaned dataset. This strengthens the interpretation by showing that the main trip-pattern differences are not limited to a small sample or a single intermediate dataset.

These findings demonstrate that unsupervised learning can reveal meaningful structure in the trip data and provide useful context for later interpretation of the full project results.

In [ ]:
```
!jupyter nbconvert --to html "/content/drive/MyDrive/COMP 333 - Project/Phas
```